9. 1

1

(3) $y = \ln x \quad (1 \le x \le 5)$

$$\int_1^5 \sqrt{1 + \frac{1}{x^2}}\, dx$$

$$= \int_1^5 \frac{\sqrt{x^2+1}}{x^2}\, dx$$

$$= \int_1^5 \frac{\sqrt{x^4+x^2}}{x^2}\, dx$$

$$= \int_1^5 \frac{\sqrt{x^2+1}}{x}\, dx$$

$$= \int_1^5 \frac{\sqrt{x^2+1}}{x^2} \cdot x\, dx$$

$\sqrt{x^2+1} = t$

$x^2+1 = t^2 \qquad x\,dx = t\,dt$

$$\int_{\sqrt{2}}^{\sqrt{26}} \frac{t^2}{t^2-1}\, dt \qquad 1 + \frac{1}{t^2-1} = 1 + \frac{1}{(t-1)(t+1)}$$

$$= \int_{\sqrt{2}}^{\sqrt{26}} 1 + \frac{1}{2}\left(\frac{1}{t-1} - \frac{1}{t+1}\right) dt$$

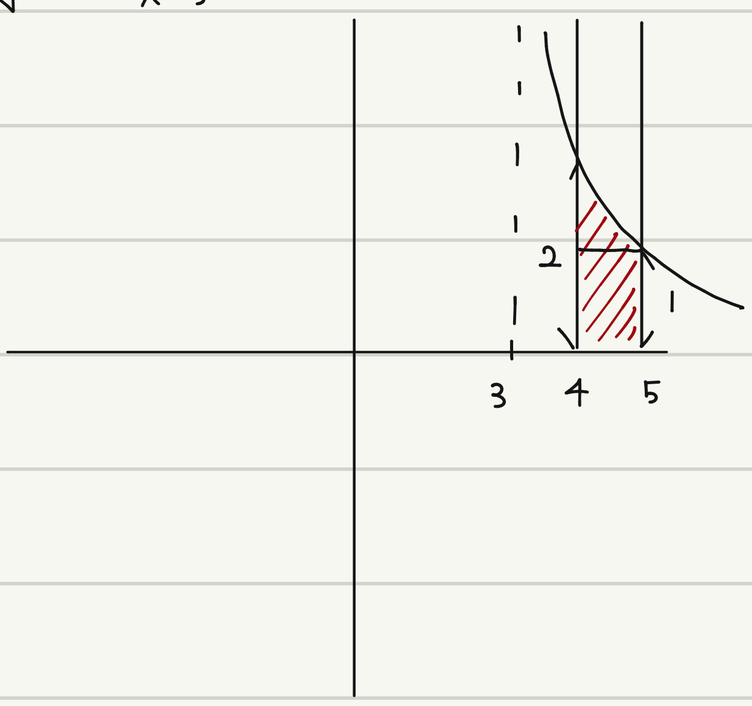
$$= \left[t + \frac{1}{2} \ln \left| \frac{t-1}{t+1} \right| \right]_{\sqrt{2}}^{\sqrt{26}}$$

$$= \sqrt{26} + \frac{1}{2} \ln \frac{\sqrt{26}-1}{1+\sqrt{26}} - \sqrt{2} - \frac{1}{2} \ln \frac{\sqrt{2}-1}{\sqrt{2}+1}$$

//

7

(4) $y = \frac{2}{x-3} \qquad y = 0, \quad x = 4, 5$



$$\int_4^5 \frac{2}{x-3}\, dx$$

$$= 2\left[\ln |x-3| \right]_4^5$$

$$= 2 \ln 2$$

//

8

(1) $y = x^3 - x^2 - 4x + 4$

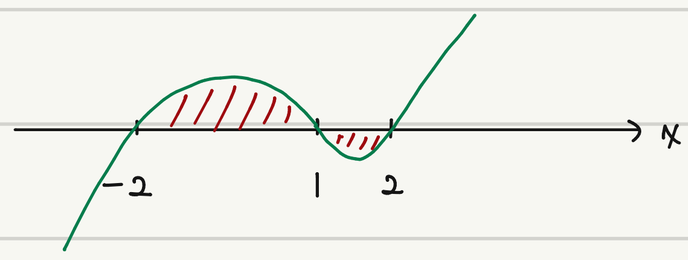
$= x^2(x-1) - 4(x-1)$

$= (x+2)(x-2)(x-1)$

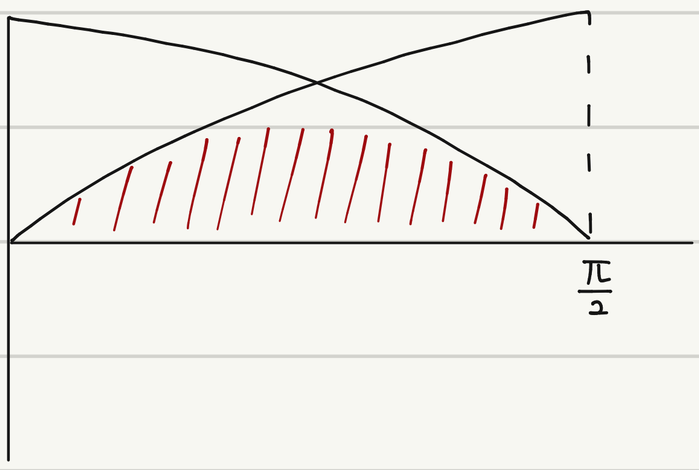
$= \frac{1}{12} \cdot 81 + \frac{1}{12}$

$= \frac{41}{6}$

//



9



$$2 \cdot \int_0^{\frac{\pi}{4}} \sin x \, dx$$

$$2 \cdot \left[-\cos x \right]_0^{\frac{\pi}{4}}$$

$$= 2\left(-\frac{\sqrt{2}}{2} + 1 \right)$$

$$= 2 - \sqrt{2} \quad //$$

10

(2) $r = \underline{6a \sin 2\theta}$ //

$$4 \int_0^{\frac{\pi}{4}} \underbrace{\frac{1}{2}}_{} \cdot 36a^2 \sin^2 2\theta \, d\theta$$

$$= 72a^2 \int_0^{\frac{\pi}{4}} \sin^2 2\theta \, d\theta$$

$$\int \sin^2 2\theta \, d\theta$$

$$= -\underbrace{\frac{1}{6}}_{} \cdot \cos^3 2\theta + C$$

$$= 72a^2 \cdot \underbrace{\frac{1}{6}}_{}$$

$$= 12a^2 \quad //$$

11 $y' = \underbrace{\frac{-2x}{2\sqrt{1-x^2}}}_{} = -\underbrace{\frac{x}{\sqrt{1-x^2}}}_{}$

(1) $y = \sqrt{1-x^2} \quad (0 \le x \le 1) \; ; \; x\frac{2}{5}$

$$\int_0^1 2\pi \cdot \sqrt{1-x^2} \cdot \sqrt{1 + \underbrace{\frac{x^2}{1-x^2}}_{}} \, dx$$

$$\underbrace{\qquad\qquad}_{1}$$

$$\underbrace{\qquad\qquad}_{1-x^2}$$

$$= 2\pi$$

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